Pain Presence, Not Location: A Causal and Information-Theoretic Analysis of Pain Assessment from Peripheral Physiology

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***Abstract*—Automated pain assessment from physiological signals has advanced rapidly, yet has focused almost entirely on detecting the presence or intensity of pain. Whether peripheral physiological signals also encode the anatomical *location* of a painful stimulus remains largely unexamined. We study this on the AI4Pain 2026 dataset, which provides electrodermal activity, blood volume pulse, respiration, and oxygen saturation for a balanced three-class problem: No Pain, Pain Arm, and Pain Hand. We build a simple, reproducible subject-independent pipeline that combines label-free per-subject normalization, interpretable features, and standard classifiers. On the official validation set our best model reaches 56.5% accuracy (95% CI 49.8–62.5) against a 41.7% challenge baseline, and 56.2% on the held-out test set, confirming that the accuracy generalizes to unseen subjects. A controlled analysis, however, traces this accuracy almost entirely to pain *detection*: No-Pain versus Pain separates cleanly, whereas Arm-vs-Hand localization stays at chance under standard features, a null that holds across four classifier families, individual subjects, a label-permutation control, and a model-independent information bound. This matches prior work on the same data, where ordinary features also fail and only a bespoke descriptor reaches a modest approximate 70%. The usable signal is carried by blood volume pulse and electrodermal activity, whereas respiration and oxygen saturation are uninformative and dilute naive fusion. Systemic autonomic responses thus index pain arousal but not stimulus site under standard peripheral features, with direct implications for physiological pain-localization systems.**

I. Introduction

Pain is an inherently subjective experience, and its reliable, objective assessment is important for patients who cannot self-report and for closing the loop in automated care [1]. Physiological signals offer an attractive route to objective assessment, as they reflect the autonomic response to a noxious stimulus and can be captured with inexpensive wearable sensors [2]. Among these, electrodermal activity (EDA) and blood volume pulse (BVP) have been studied most, alongside respiration and cardiovascular measures [2], [3]. Existing studies can be broadly grouped into three directions: hand-crafted physiological features with classical classifiers [2], end-to-end deep models on raw or spectral representations [4], [5], and multimodal fusion of several signals [6]. Comparatively, almost all of this work addresses a single question: whether the subject is in pain, and how intense it is. The related but distinct question of *where* the pain is, the anatomical location of the stimulus, has received little attention.

In this study, we focus on subject-independent pain assessment on the AI4Pain 2026 dataset, which poses both questions jointly through a balanced three-class task: No Pain, Pain Arm, and Pain Hand. Our pipeline can be split into three stages: 1) label-free per-subject normalization that removes inter-individual offsets without using labels or cross-subject information; 2) interpretable per-segment physiological features, including EDA tonic and phasic descriptors, BVP pulse-rate variability, respiration rate, and summary statistics; and 3) a standard classifier evaluated under leave-one-subject-out cross-validation, with a random forest as a simple reference and gradient-boosted trees as the best of four families. The design is simple and reproducible, and it already improves over the challenge baseline.

The specific difficulty we encountered is that the two pain classes are almost inseparable: while No Pain versus Pain is recovered with high accuracy, the Arm versus Hand distinction stays at chance. We therefore propose to treat the localization question as the primary object of analysis rather than to claim it away, and we verify the effect with a label-permutation control, a per-subject significance test, a hierarchical model, and a per-modality ablation. Based on that, while our system outperforms the baseline on the official metric, we show that

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this accuracy is driven by detection and that localization is at chance. In summary, the main contributions of this study are as follows:

- We decompose the task into detection and localization and show that all achievable accuracy comes from detection, while Arm-vs-Hand localization stays at chance across four classifier families, individual subjects, a label-permutation control, and a within-subject test. Convergent causal effect-size and information-theoretic evidence (approximate 0 bits about site vs. 0.21 about pain) confirms the site is undecodable from standard features, so the benchmark's three-class accuracy should count as a detection score.
- Ablation explains why detection rests on blood volume pulse and electrodermal activity, while respiration and oxygen saturation are near-chance and dilute naive fusion; the only site-related signal is a weak respiration-variability effect below the classification threshold.

II. Related Work

Objective pain assessment is motivated by patients who cannot self-report, and the field has been surveyed extensively [1], [7]. Physiology is attractive since a noxious stimulus triggers a stereotyped autonomic response, sympathetic activation and parasympathetic withdrawal, that is measurable in heart- rate variability [8], [9] and electrodermal activity. This mechanism is tied to pain magnitude and arousal, which already hints that *location* may be hard to recover from such peripheral signals. By contrast, central measures can carry far richer pain information: an fMRI signature predicts thermal-pain intensity within individuals [10], underscoring that what is decodable depends strongly on the signal source.

On benchmarks such as BioVid [3], and in the broader wearable-affect setting exemplified by WESAD [11], methods range from hand-crafted biopotential features with SVMs [2], [12] to deep temporal networks: denoising convolutional autoencoders [13], 2-D CNNs over fused signals [4], multiscale CNN–transformer encoders [5], and EDA-specific recurrent models [14]. Personalization via multi-task learning has also been used to handle individual differences [15].

The AI4Pain series moved from fNIRS and facial video [16], [17] to peripheral physiological signals [18]; recent entries fuse EDA and cardiovascular signals with cross-attention transformers [6] or exploit single modalities such as respiration [19]. Most of these target pain *presence/intensity*; the next-closest classifies pain *type* (social vs. physical) [20], not anatomical site. The exception, and the closest prior work to ours, is Aziz et al. [21], they report a two-stage EDA pipeline that detects pain at 77.9% and then localizes it (hand vs. forearm) at 69.7% under LOSO. That above-chance localization, however, emerges only from a specialized engineered descriptor with feature fusion and selection; *standard* time-domain EDA features leave the task at chance in their own experiments, and an earlier study reports 56%. Their standard-feature result thus agrees with ours, and we revisit their finding directly below.

Cross-subject generalization is the recurring obstacle: models degrade on unseen subjects, which is why subject-independent (LOSO) protocols, per-subject normalization [2], subject-invariant contrastive learning [22], and, increasingly, self-supervised foundation models for physiological signals [23], [24] are used. We articulate the gap in three parts: (i) physiological pain work overwhelmingly targets presence/intensity, and whether systemic autonomic signals encode the *site* of a stimulus remains unsettled, as the one prior attempt on this kind of data reaches 70% only with a bespoke descriptor while standard features stay at chance [21]; (ii) the two AI4Pain 2026 sites lie on the same limb, making any peripheral difference small; and (iii) the autonomic response is mechanistically tied to pain magnitude rather than spatial origin. We therefore test for site information directly, using classification, causal effect sizes, and information-theoretic bounds [25], [26], rather than assuming a flat three-class task is uniformly learnable.

III. Method

This section describes the key components of our system in detail. The proposed system can be split into three main parts: per-subject normalization, feature extraction, and classification, followed by the detection/localization analysis.

A. Problem setup and notation

For subject $s \in \{1, \dots, S\}$, each around 10-second segment is a multichannel window $\mathbf{x} \in \mathbb{R}^{C \times T}$ with $C = 4$ channels (EDA, BVP, Resp, $\mathrm{SpO_2}$) sampled at $100\,\mathrm{Hz}$, and a label $y \in \{\textsc{NoPain}, \textsc{Arm}, \textsc{Hand}\}$. The goal is a subject-independent classifier $f : \mathbf{x} \mapsto y$ that generalizes to unseen subjects; we therefore develop and select models under leave-one-subject-out (LOSO) cross-validation and report on a disjoint validation set. We also study two sub-problems: *detection* (NoPain vs. Pain$= \{$Arm, Hand$\}$) and *localization* (Arm vs. Hand).

B. Per-subject normalization

For each subject s and channel c we z-score using statistics computed only from that subject's own segments, $\tilde{x}_{c,t} = (x_{c,t} - \mu_{s,c})/\sigma_{s,c}$, where $\mu_{s,c}, \sigma_{s,c}$ are the mean and standard deviation over all of subject s's samples in channel c. Since these statistics use neither labels nor any other subject, the transform introduces no leakage across the subject-disjoint splits, while removing the dominant inter-individual offsets. It is *transductive*: it assumes batch access to a subject's segments at inference to estimate $\mu_{s,c}, \sigma_{s,c}$, which holds for the per-subject challenge test but should be noted as a deployment assumption.

C. Feature extraction

Each segment yields a length-invariant feature vector per channel: generic statistics (mean, dispersion, slope, RMS, skewness, and kurtosis), Hjorth descriptors, Welch spectral descriptors, and within-window dynamics. To these we add signal-specific descriptors, namely EDA tonic level and phasic skin-conductance responses, BVP pulse rate and its heart-rate-variability proxies SDNN, RMSSD, and pNN50, and respiration rate and depth, together with pairwise inter-channel correlations, for 136 features in total. The EDA tonic/phasic separation follows the standard decomposition view [27].

D. Classification

We map the feature vector to one of the three classes and compare three classifiers, namely a random forest, an RBF-kernel support vector machine, and gradient-boosted trees (LightGBM), as well as an end-to-end 1-D convolutional network on the raw windows. The choice is dictated by the data regime rather than a preference for simplicity: with only 41 training subjects and a 136-dimensional table of handcrafted features, tree ensembles are low-variance, robust learners on tabular inputs and remain the standard, competitive choice for pain-from-physiology work [2], [12], whereas the RBF-SVM and the raw-signal CNN test whether a kernel method or an end-to-end model helps at this scale. We additionally evaluate a hierarchical variant that first separates No Pain from Pain and then Arm from Hand, to isolate the localization decision.

E. Analysis methods

Beyond accuracy, we quantify how much each condition is detectable and how strongly it is caused. As stimulation site and intensity are counterbalanced within subject [28], the within-subject paired difference of a (standardized) feature between two conditions approximates a causal average treatment effect; we summarize it by Cohen's $d_z = \bar{d}/\mathrm{sd}(d)$ over subjects and test it with a one-sample t-test, controlling the false-discovery rate (Benjamini–Hochberg) across the 136 features. To bound decodability independently of any particular model, we estimate the *usable information* [25] $I_V(\mathbf{x}\to y) = H(y) - \mathrm{CE}$, where $H(y)$ is the label entropy and CE is the cross-entropy (in bits) of a calibrated predictor under subject-grouped cross-validation; I_V is clamped at 0 when the predictor does not beat the marginal. A label-permutation null calibrates the floor.

IV. Experiments

A. Dataset and metric

AI4Pain 2026 [28] provides EDA, BVP, respiration, and SpO_2 at 100 Hz from 65 participants (23 female; mean age 29.1 ± 8.3 years), split subject-disjointly into 41/12/12 train/validation/test, with 12 balanced 10-second segments per class per subject. Pain is evoked by transcutaneous electrical nerve stimulation (TENS) at the forearm (Arm) and the back of the hand (Hand), with site and intensity counterbalanced. The full acquisition protocol, including electrode placement, stimulus intensity, and inter-stimulus timing, follows the organizers' documentation [1], [21], [28]. The No-Pain class combines baseline and post-stimulus rest windows. Following the challenge, the primary metric is overall accuracy; we additionally report a detection accuracy (No-Pain vs. Pain) and a localization accuracy (Arm vs. Hand among detected-pain segments). With balanced classes, macro-F1 tracks accuracy closely (e.g. 0.567 vs. 0.565 for the best model), so we report accuracy throughout.

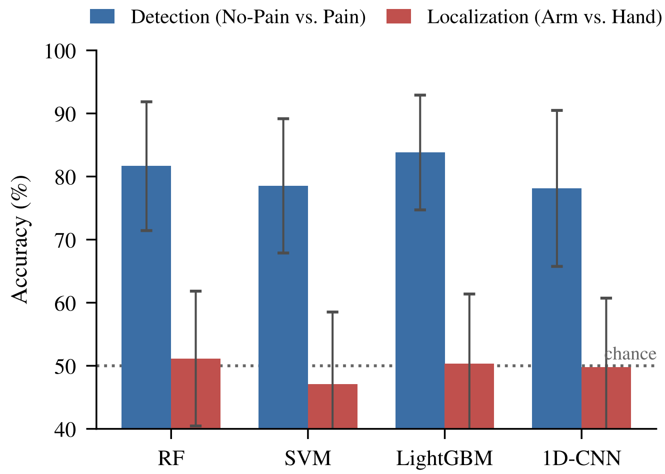

Fig. 1. Detection vs. localization across classifier families (LOSO, mean±std over 41 subjects). Every model detects pain well (∼80%) but classifies the stimulation site at chance (∼50%, dotted line).

TABLE I
SUBJECT-INDEPENDENT PAIN CLASSIFICATION ON AI4PAIN 2026 (OUR 136 PHYSIOLOGICAL FEATURES + PER-SUBJECT NORMALIZATION). ACC. IS THE OVERALL THREE-CLASS (NO-PAIN/ARM/HAND) ACCURACY: THE OFFICIAL CHALLENGE METRIC, NOT AN AVERAGE OF THE TWO SUB-TASKS; DET. = NO-PAIN VS PAIN; LOC. = ARM VS HAND AMONG DETECTED PAIN (CHANCE 50). BASELINES ARE ORGANIZER-REPORTED [28].

Method	LOSO Acc.	val Acc.↑	val Det.	val Loc.
SVM (multimodal, raw) †	–	41.7	–	–
SVM (BVP, raw) †	–	43.8	–	–
Nearest centroid (floor)	52.7 ± 8.9	53.5	81.5	50.6
RandomForest	55.1 ± 9.9	54.9	83.1	51.8
SVM-RBF	49.8 ± 10.4	55.1	82.6	52.8
LightGBM	55.9 ± 9.5	56.5	84.7	52.9
1D-CNN (raw)	52.1 ± 9.9	52.1	79.6	50.8

† organizer baseline.

B. Implementation

Each segment is described by 136 features across the four channels. Model selection uses LOSO cross-validation on the 41 training subjects, where gradient-boosted trees are best; we then report each configuration on the validation set with a subject-level bootstrap CI. Feature-set development was checked against validation at a few iterations, so the validation estimate is slightly optimistic; the conclusions are unaffected, since the LOSO ranking agrees and the result is detection-driven throughout.

C. Results and discussion

Table I reports the main results, and Table II the confusion matrix. All of our models improve over the baseline on the official metric, with gradient-boosted trees best on LOSO; on validation the inter-model gaps fall within the ±6.5-pt bootstrap CI and are not statistically distinguishable (Fig. 1). On the held-out test set (12 subjects, labels withheld), our best model achieves 56.2%, essentially matching its

TABLE II

CONFUSION MATRIX (RANDOMFOREST REFERENCE MODEL, SUMMED OVER LOSO FOLDS). THE ARM AND HAND ROWS ARE NEAR-IDENTICAL: PAIN IS SEPARATED FROM NO-PAIN, BUT THE TWO PAIN LOCATIONS ARE NOT.

True \ Pred	No Pain	Arm	Hand
No Pain	399	60	33
Arm	87	217	188
Hand	85	192	215

56.5% validation accuracy and indicating that the val-selected pipeline generalizes to unseen subjects. On LOSO, LightGBM significantly exceeds both a nearest-centroid floor and the end-to-end 1-D CNN, the weak CNN being consistent with the small number of training subjects, whereas the between-model gap on localization is, as expected, not significant ($p = 0.81$). Most of the improvement over the organizer baseline comes from features and per-subject normalization rather than the classifier: LightGBM sits only approximate 3 pts above the nearest-centroid floor, and normalization alone adds 3.9 pts (Table IV), consistent with interpreting the metric as detection. The decomposition is the key observation: pain detection is high, whereas Arm-vs-Hand localization is at chance. The confusion matrix makes this explicit, as the Arm and Hand rows are near-identical. A label-permutation control collapses both three-class and Arm-vs-Hand accuracy to chance, confirming the detection signal is genuine and the localization result is not an artifact. A per-subject test finds only 3 of 41 subjects above chance on localization (consistent with the false-positive rate), and a hierarchical model does not improve over the flat classifier. More directly, a within-subject Arm-vs-Hand classifier (trained and tested on the same subject) is also at chance ($50.5 \pm 14.2\%$), so the failure is not merely due to inter-individual variability: site is not decodable even within-subject, though the per-subject sample ($\sim$24 segments) is small, so we say site is not decodable at this sample size and protocol rather than provably absent. The only hints of a site-related effect are a weak difference in respiration variability and in BVP amplitude decay; these reach significance only under narrow, pre-specified tests, do not survive correction across all features, and are far too small to support classification.

D. Localization analysis

The organizers report counterbalancing stimulation site and intensity "to avoid confounding factors due to order effects" [28], so within-subject contrasts approximate causal average treatment effects (ATE) of the manipulated condition rather than mere associations, subject to the caveats in Sec. IV-H: the design is counterbalanced rather than fully randomized, and the No-Pain class mixes rest windows. The pain effect is large and pervasive: 65 of 136 standardized features shift significantly (FDR<0.05), up to $|d_z| = 1.6$, with BVP amplitude and variability falling as expected under vasoconstriction. In contrast, the site effect is negligible, as no feature survives correction across all features ($|d_z| \leq 0.42$). The nociceptive stimulus thus reliably changes autonomic state, whereas the stimulation site does not, which restates why detection succeeds and localization does not. This is experimental evidence in healthy adults under TENS-evoked acute pain, not clinical evidence (Sec. IV-H).

To show the localization failure is intrinsic to the signals rather than to our models, we estimate the usable information $I_V(X \to Y) = H(Y) - \mathrm{CE}$ a flexible predictor can extract under subject-grouped cross-validation [25]. I_V is a non-negative (clamped) estimate: for site the model's cross-entropy exceeds $H(Y)$, i.e. it does not beat the marginal, so I_V floors at 0. The features carry 0.21 bits about pain presence (23% of the label entropy) but 0.00 bits about stimulation site (Fig. 2; a label-permutation null gives 0.00 for both; Bayes-error proxy 0.50 for site). The site value is not merely small but at the floor: the unclamped estimate is *negative* ($H(Y) - \mathrm{CE} = -0.33$ bits, i.e. the predictor is 0.33 bits worse than the marginal). Within this estimator and sample, the stimulation site is therefore (near-)undecodable from these peripheral signals under the standard feature representation we test. A stronger classifier would not help; these features carry the information at the floor. The scope matters here: a descriptor engineered specifically for localization has reached $\sim$70% on related data [21] (Sec. IV-H), so the limit is one of representation, not a universal impossibility. There is no paradox in that $\sim$70%: usable information is a property of a representation, not of the raw signal, so a purpose-built descriptor can expose a faint site cue that generic autonomic features do not. Our claim is scoped accordingly: these standard features, not the EDA signal itself, lack decodable site information; and, since the shortfall is representational rather than one of classifier capacity, a higher-capacity classifier over the same features cannot recover what they do not encode, consistent with the within-subject and four-family results above.

Motivated by the differing afferent conduction distance and local vasomotor response of the two sites, we engineered features targeting localization specifically: response-onset timing, BVP pulse morphology and amplitude-decay (vasoconstriction) kinetics, and the EDA–BVP cross-modal lag. None separated Arm from Hand above chance, in either within- or cross-subject classification. The strongest individual cue, a consistent BVP amplitude-decay difference (vasoconstriction kinetics), did not survive multiple-comparison correction. We report this negative attempt as it bounds what is recoverable: the site signal is not merely hard to model but largely absent from these peripheral signals under this protocol.

E. Ablations

A per-modality ablation (Table III) shows BVP and EDA carry the detectable signal, while respiration and SpO_2 are near chance and their removal does not reduce detection, indicating that naive fusion of all four signals is diluted by the weak modalities; the most informative individual features are BVP pulse amplitude/variability and EDA tonic descriptors (Fig. 3), the canonical sympathetic-arousal markers. Label-free per-subject normalization is in turn the single most important

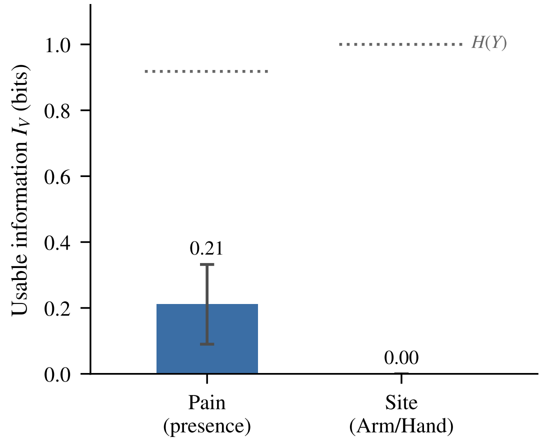

Fig. 2. Usable information I_V (bits) the features carry about pain presence vs. stimulation site (dotted lines: label entropy $H(Y)$). Pain 0.21 bits; site 0.00 bits (clamped; see text).

TABLE III
PER-MODALITY ABLATION (RANDOMFOREST REFERENCE MODEL, LOSO ACCURACY, %). SINGLE MODALITIES (TOP) AND LEAVE-ONE-MODALITY-OUT (BOTTOM). BVP AND EDA CARRY DETECTION; REMOVING RESP OR SpO_2 BARELY CHANGES IT; LOCALIZATION STAYS AT CHANCE THROUGHOUT.

Modalities	Acc.	Det.	Loc.
BVP	52.1	79.1	50.4
EDA	50.1	74.2	50.2
Resp	33.9	54.3	51.6
SpO_2	36.2	56.8	48.6
all − BVP	51.6	74.0	53.2
all − EDA	52.9	78.9	52.3
all − Resp	54.6	81.8	50.5
all − SpO_2	55.8	82.0	52.3

design choice (Table IV): removing it drops LightGBM from 55.9 to 52.0% LOSO accuracy and detection from 83.8 to 77.9%, confirming that suppressing inter-individual offsets, not model capacity, is what most aids cross-subject generalization.

F. Error analysis

The error budget (Table II, RandomForest reference model) has two distinct modes. The dominant one is Arm↔Hand confusion: among segments correctly identified as painful, true-Arm windows are labeled Hand 46.4% of the time and true-Hand windows are labeled Arm 47.2%, a near-symmetric, near-chance split that mirrors the absence of a site signal rather than a directional bias. The secondary mode is the detection boundary: 17.5% of pain segments are missed (predicted No-Pain) and 18.9% of No-Pain segments raise a false alarm. These false alarms concentrate where the protocol is weakest: No-Pain mixes initial-baseline and post-stimulus rest windows, and residual sympathetic arousal in the rest windows resembles mild pain. The released labels do not separate the two subtypes, but using segment index as a proxy, since baseline windows precede rest windows in the protocol, the false-alarm rate is higher on putative rest windows than on putative

TABLE IV
EFFECT OF LABEL-FREE PER-SUBJECT NORMALIZATION (LIGHTGBM). IT IS THE SINGLE MOST IMPORTANT DESIGN CHOICE FOR CROSS-SUBJECT GENERALIZATION.

Normalization	LOSO Acc.	LOSO Det.	val Acc.
With (ours)	55.9 ± 9.5	83.8	56.5
Without	52.0 ± 9.6	77.9	54.4

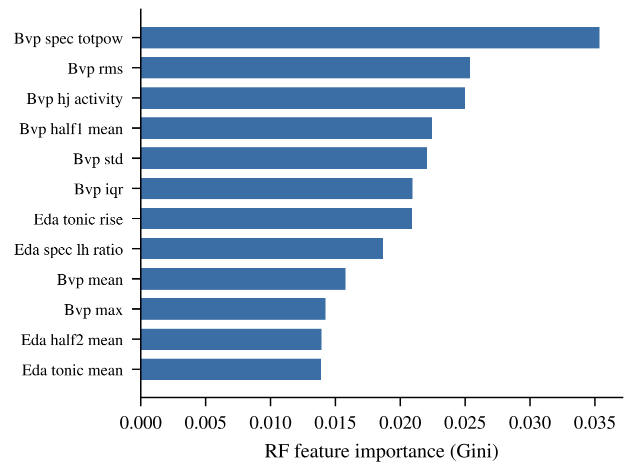

Fig. 3. Top random-forest feature importances. The most informative features are BVP amplitude/variability and EDA tonic descriptors (sympathetic-arousal markers), consistent with the modality ablation.

baseline windows (22.8% vs. 15.9%; permutation $p = 0.068$), a trend consistent with, though not conclusive proof of, the residual-arousal confound; detection on clean baseline windows is correspondingly stronger (cf. Sec. IV-H). Detection accuracy also varies markedly across subjects (0.50–0.97, mean 0.82 ± 0.10), indicating that per-subject normalization removes much, but not all, inter-individual variability. No error mode is specific to Arm vs. Hand beyond chance, so the residual error is dominated by the irreducible localization confusion plus a smaller, partly protocol-induced detection error.

G. Implications

Our primary, transferable contribution is methodological rather than a leaderboard number: on a benchmark that mixes an easy and a hard sub-task into one accuracy, that accuracy can be a detection score in disguise, and only a decomposition reveals it. Several practical points follow. First, for this benchmark the three-class accuracy is, to a first approximation, a relabeled pain-detection score; entries and leaderboards should therefore report the detection and localization components separately, since a higher three-class number can reflect better detection while localization stays at chance. Second, the convergent evidence from a paired causal effect-size analysis, which finds a negligible site effect, and a model-independent usable-information bound of ≈ 0 bits for site indicates that, within the standard autonomic feature families we test, a more expressive classifier is unlikely to help: the limitation is one of representation, not classifier capacity. This is not a claim of universal impossibility, since a descriptor engineered specifically for localization has reached $\sim$70% on related data [21]; rather, it indicates that recovering site needs a different representation (or signal), not merely a stronger model. Third, recovering stimulus site plausibly requires signals with spatial specificity, such as surface EMG near the stimulation site, regional skin-conductance, or imaging, rather than systemic autonomic channels alone. Finally, a task-design note: in many clinical settings, such as ICU sedation and post-operative monitoring, whether a patient is in pain is the actionable output while localization is secondary; combined with the anatomical proximity of the two AI4Pain sites (same limb, shared afferent pathways), this suggests future peripheral-sensor benchmarks should target anatomically distinct sites, or weight the detection and localization components by their differing clinical value. These observations transfer to any pain-localization effort built on peripheral physiology.

H. Limitations

Our localization result is a negative result for the Arm/Hand distinction; it characterizes the present sensors, sample size, standard feature representation, and protocol, and does not prove localization is impossible in principle. In particular, Aziz et al. [21] reach approximate 70% on the same hand/forearm contrast using a bespoke multi-domain EDA descriptor with feature fusion and selection; our null is therefore specific to standard autonomic features, and reproducing their specialized descriptor under our leakage-safe, no-peek protocol is left to future work. The two stimulation sites lie on the same limb, which may make them especially hard to separate. The treatment-effect interpretation rests on the organizers' counterbalanced design; we did not control stimulus order ourselves, and the No-Pain class combines baseline with post-stimulus rest windows, so residual autonomic arousal in the rest windows, along with any anticipatory arousal if participants sense an impending stimulus, may inflate the detection effect; the pain effect size should be interpreted with this confound in mind, whereas the site contrast, the focus of this paper, is unaffected. Validation uses 12 subjects, so the held-out estimate is wide (CI 49.8–62.5); stochastic models use a single seed (variance is across subjects). Finally, the cohort is young and male-skewed and we did not analyze per-sex or per-age fairness; this should be examined before any deployment claim.

V. Conclusion

This paper asked whether peripheral physiological signals localize pain or only detect it. On AI4Pain 2026, our subject-independent pipeline improves three-class accuracy over the challenge baseline, but a controlled analysis attributes the entire gain to detection, with Arm-vs-Hand localization at chance under standard physiological features. Causal effect sizes and a model-independent information bound concur: systemic autonomic responses index pain arousal, not stimulus site. Recovering site therefore requires signals with spatial specificity, such as site-local or neuromuscular measurements, or representations engineered for localization [21], rather than standard autonomic features alone.

Ethical Impact Statement

The dataset was collected by the challenge organizers (University of Canberra) under their institutional human-research ethics approval with written informed consent, following the Declaration of Helsinki [1], [28]. We conducted a secondary analysis of the de-identified challenge release and at no point attempted re-identification or linkage to external records. The signals are peripheral physiological recordings from healthy adult volunteers under an experimentally evoked nociceptive stimulus, not observations of clinical pain.

This study is a methodological benchmark analysis, not a clinical or diagnostic tool, and its results must not be used to infer pain, adjust analgesia, or make care decisions for any individual. Automated pain assessment carries a specific risk of displacing patient self-report, and a system trusted beyond its evidence could withhold treatment from people in genuine pain or subject them to unwarranted monitoring. Our findings sharpen rather than soften this concern: we show that peripheral physiology on this dataset indexes the presence of pain but not its location, so any downstream use must not read a detection signal as fine-grained localization. We keep every claim proportional to the evidence and report the negative localization result plainly to discourage such over-reading.

The cohort is small, young, and male-skewed, and we did not perform a per-sex or per-age fairness analysis; performance may therefore not transfer across demographic groups, clinical populations, or pain of non-experimental origin. Any move toward real-world use would require validation on a larger, more representative and clinically relevant sample, an explicit fairness audit, and human oversight, none of which this work provides or claims. The models are lightweight classical classifiers and a small 1D-CNN, with negligible computational and environmental cost.

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References

[1] R. Fernandez Rojas, N. Hirachan, N. Brown, G. Waddington, L. Murtagh, B. Seymour, and R. Goecke, "Multimodal physiological sensing for the assessment of acute pain," *Frontiers in Pain Research*, vol. 4, p. 1150264, 2023.

[2] F. Pouromran, S. Radhakrishnan, and S. Kamarthi, "Exploration of physiological sensors, features, and machine learning models for pain intensity estimation," *Plos one*, vol. 16, no. 7, p. e0254108, 2021.

[3] S. Walter, S. Gruss, H. Ehleiter, J. Tan, H. C. Traue, P. Werner, A. Al-Hamadi, S. Crawcour, A. O. Andrade, and G. M. Da Silva, "The BioVid heat pain database: Data for the advancement and systematic validation of an automated pain recognition system," in *2013 IEEE international conference on cybernetics (CYBCO)*. IEEE, 2013, pp. 128–131.

[4] P. Thiam, H. Hihn, D. A. Braun, H. A. Kestler, and F. Schwenker, "Multi-modal pain intensity assessment based on physiological signals: A deep learning perspective," *Frontiers in Physiology*, vol. 12, p. 720464, 2021.

[5] Z. Lu, B. Ozek, and S. Kamarthi, "Transformer encoder with multi-scale deep learning for pain classification using physiological signals," *Frontiers in Physiology*, vol. 14, p. 1294577, 2023.

[6] J. Farmani, G. Bargshady, S. Gkikas, M. Tsiknakis, and R. F. Rojas, "A CrossMod-Transformer deep learning framework for multi-modal pain detection through EDA and ECG fusion," *Scientific Reports*, vol. 15, no. 1, p. 29467, 2025.

[7] P. Werner, D. Lopez-Martinez, S. Walter, A. Al-Hamadi, S. Gruss, and R. W. Picard, "Automatic recognition methods supporting pain assessment: A survey," *IEEE Transactions on Affective Computing*, vol. 13, no. 1, pp. 530–552, 2019.

[8] J. Koenig, M. N. Jarczok, R. J. Ellis, T. Hillecke, and J. F. Thayer, "Heart rate variability and experimentally induced pain in healthy adults: a systematic review," *European journal of pain*, vol. 18, no. 3, pp. 301–314, 2014.

[9] F. Shaffer and J. P. Ginsberg, "An overview of heart rate variability metrics and norms," *Frontiers in public health*, vol. 5, p. 290215, 2017.

[10] T. D. Wager, L. Y. Atlas, M. A. Lindquist, M. Roy, C.-W. Woo, and E. Kross, "An fMRI-based neurologic signature of physical pain," *New England Journal of Medicine*, vol. 368, no. 15, pp. 1388–1397, 2013.

[11] P. Schmidt, A. Reiss, R. Duerichen, C. Marberger, and K. Van Laerhoven, "Introducing wesad, a multimodal dataset for wearable stress and affect detection," in *Proceedings of the 20th ACM international conference on multimodal interaction*, 2018, pp. 400–408.

[12] S. Gruss, R. Treister, P. Werner, H. C. Traue, S. Crawcour, A. Andrade, and S. Walter, "Pain intensity recognition rates via biopotential feature patterns with support vector machines," *PloS one*, vol. 10, no. 10, p. e0140330, 2015.

[13] P. Thiam, H. A. Kestler, and F. Schwenker, "Multimodal deep denoising convolutional autoencoders for pain intensity classification based on physiological signals." in *ICPRAM*, 2020, pp. 289–296.

[14] F. Pouromran, Y. Lin, and S. Kamarthi, "Personalized deep bi-lstm rnn based model for pain intensity classification using eda signal," *Sensors*, vol. 22, no. 21, p. 8087, 2022.

[15] D. Lopez-Martinez and R. Picard, "Multi-task neural networks for personalized pain recognition from physiological signals," in *2017 Seventh International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW)*. IEEE, 2017, pp. 181–184.

[16] R. Fernandez-Rojas, C. Joseph, N. Hirachan, B. Seymour, and R. Goecke, "The AI4Pain grand challenge 2024: Advancing pain assessment with multimodal fNIRS and facial video analysis," in *2024 12th International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW)*. IEEE, 2024, pp. 55–60.

[17] S. Gkikas and M. Tsiknakis, "Twins-PainViT: Towards a modality-agnostic vision transformer framework for multimodal automatic pain assessment using facial videos and fNIRS," in *2024 12th International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW)*. IEEE, 2024, pp. 13–21.

[18] R. Fernandez-Rojas, C. Joseph, N. Hirachan, B. Seymour, and R. Goecke, "The AI4Pain grand challenge 2025: Advancing pain assessment with multimodal physiological signals," in *Companion Proceedings of the 27th International Conference on Multimodal Interaction*, 2025, pp. 147–152.

[19] S. Gkikas, I. Kyprakis, and M. Tsiknakis, "Efficient pain recognition via respiration signals: A single cross-attention transformer multi-window fusion pipeline," in *Companion Proceedings of the 27th International Conference on Multimodal Interaction*, 2025, pp. 70–79.

[20] E.-H. Jang, Y.-J. Eum, D. Yoon, and S. Byun, "Classifying social and physical pain from multimodal physiological signals using machine learning," *Scientific Reports*, vol. 15, no. 1, p. 27674, 2025.

[21] S. Aziz, C. Joseph, N. Hirachan, L. Murtagh, G. Chetty, R. Goecke, and R. Fernandez-Rojas, "A two-stage architecture for identifying and locating the source of pain using novel multi-domain binary patterns of eda," *Biomedical Signal Processing and Control*, vol. 104, p. 107454, 2025.

[22] J. Y. Cheng, H. Goh, K. Dogrusoz, O. Tuzel, and E. Azemi, "Subject-aware contrastive learning for biosignals," *arXiv preprint arXiv:2007.04871*, 2020.

[23] A. Pillai, D. Spathis, F. Kawsar, and M. Malekzadeh, "Papagei: Open foundation models for optical physiological signals," in *International Conference on Learning Representations*, vol. 2025, 2025, pp. 48 230–48 261.

[24] M. Saha, M. A. Xu, W. Mao, S. Neupane, J. M. Rehg, and S. Kumar, "Pulse-PPG: An open-source field-trained PPG foundation model for wearable applications across lab and field settings," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 9, no. 3, pp. 1–35, 2025.

[25] Y. Xu, S. Zhao, J. Song, R. Stewart, and S. Ermon, "A theory of usable information under computational constraints," in *Int. Conf. on Learning Representations (ICLR)*, 2020.

[26] M. I. Belghazi, A. Baratin, S. Rajeshwar, S. Ozair, Y. Bengio, A. Courville, and D. Hjelm, "Mutual information neural estimation," in *International conference on machine learning*. PMLR, 2018, pp. 531–540.

[27] A. Greco, G. Valenza, A. Lanata, E. P. Scilingo, and L. Citi, "cvxeda: A convex optimization approach to electrodermal activity processing," *IEEE transactions on biomedical engineering*, vol. 63, no. 4, pp. 797–804, 2015.

[28] R. Fernandez-Rojas, C. Joseph, N. Hirachan, B. Seymour, and R. Goecke, "The ai4pain grand challenge 2026: A multimodal physiological dataset and benchmark for pain detection and localisation." in *2026 14th International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW)*. IEEE, 2026.